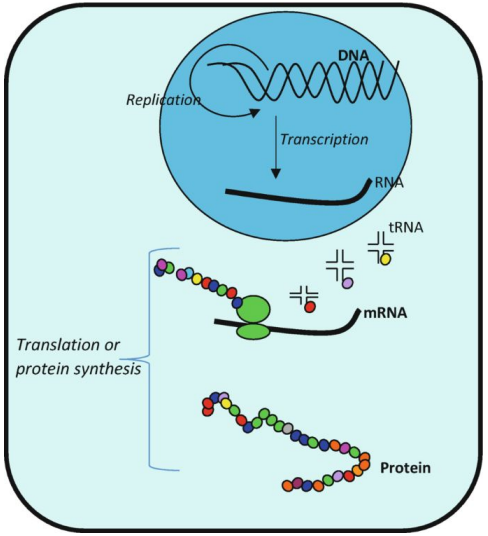
BME 105: Genetics in the Genomics Era

Lecture 12: How do protein coding and non-coding mutations affect genes?



Instructor: Shelbi Russell, PhD

Course website on canvas.ucsc.edu

Lecture 12 Learning Objectives

- Deepen your knowledge of transcription and translation (the central dogma of molecular biology)
- Understand the key differences between bacterial and eukaryotic gene expression:
  - Nuclear organelle separates transcription from translation
  - Introns need to be spliced out of eukaryotic pre-mRNA, and 5’-caps and 3’-A tails need to be added
  - Eukaryotic chromatin can be compacted around nucleosomes
- Understand how mutations change phenotypes
  - Coding DNA vs non-coding DNA
  - Silent vs missense vs nonsense vs frameshift
  - Loss of function - amorphic, hypomorphic
    - Usually recessive, unless haploinsufficient
  - Gain of function - hypermorphic, neomorphic, antimorphic (dominant-negative)
    - Usually dominant
  - Suppressor mutations

Central dogma of molecular biology

Useful supplemental videos for this lecture:

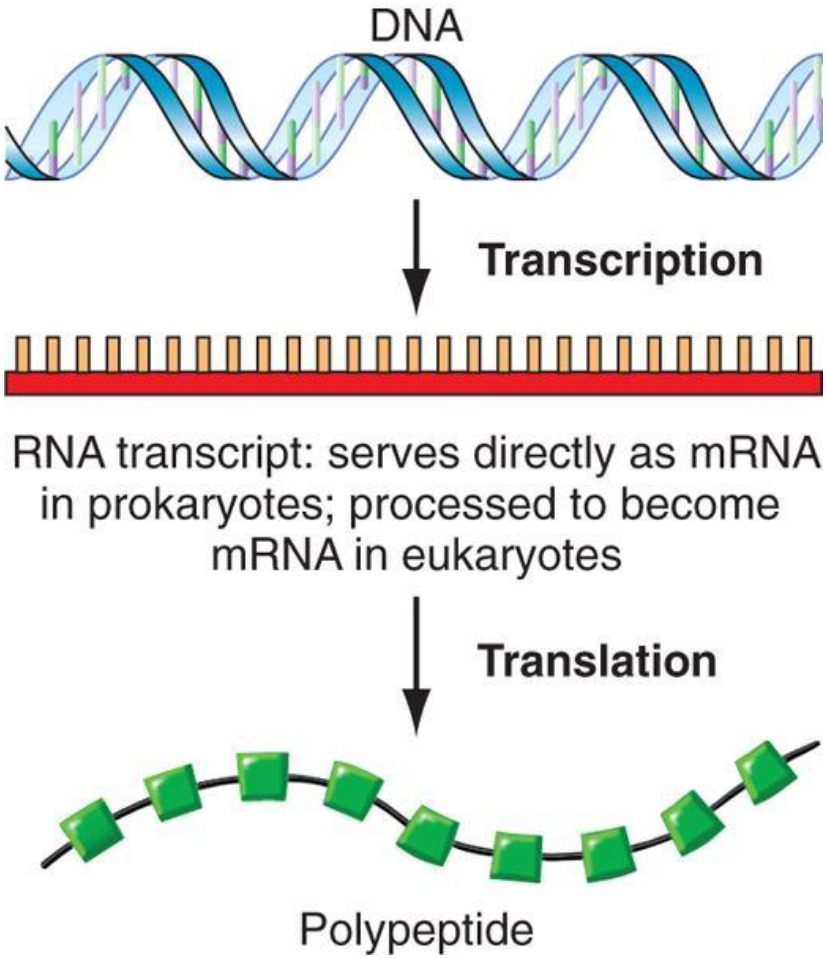
Central Dogma:
https://www.jove.com/science-education/11479/the-central-dogma

Gene expression:
https://www.jove.com/science-education/10797/what-is-gene-expression

Transcription:
https://www.jove.com/science-education/10794/transcription

RNA Splicing:
https://www.jove.com/science-education/11587/rna-splicing

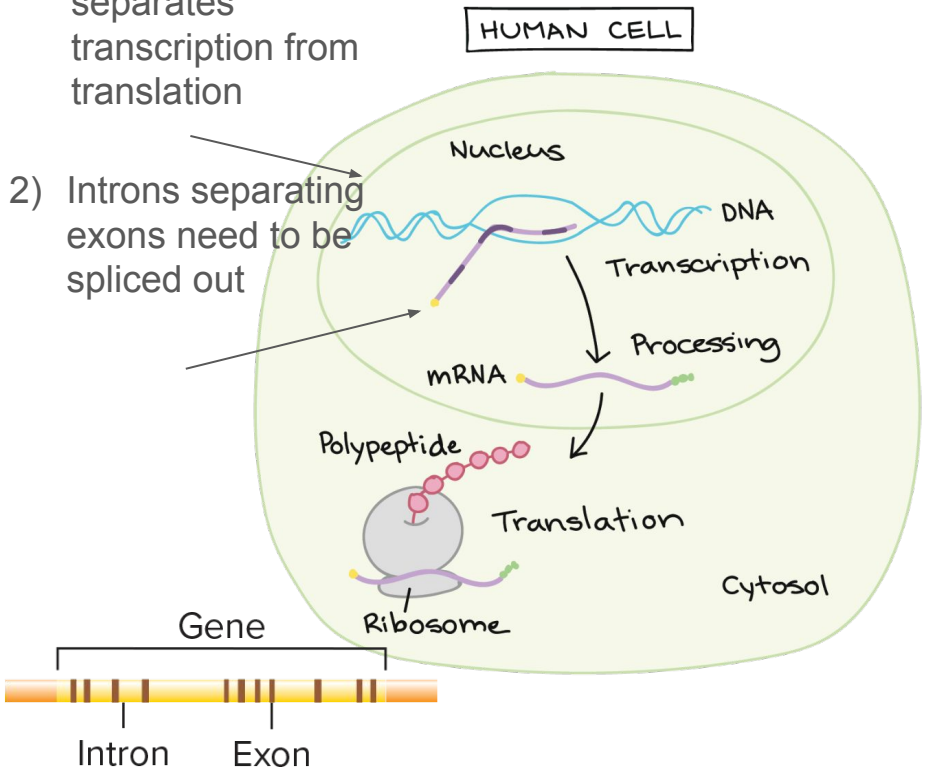
Protein synthesis/translation:
https://www.jove.com/science-education/11594/translation

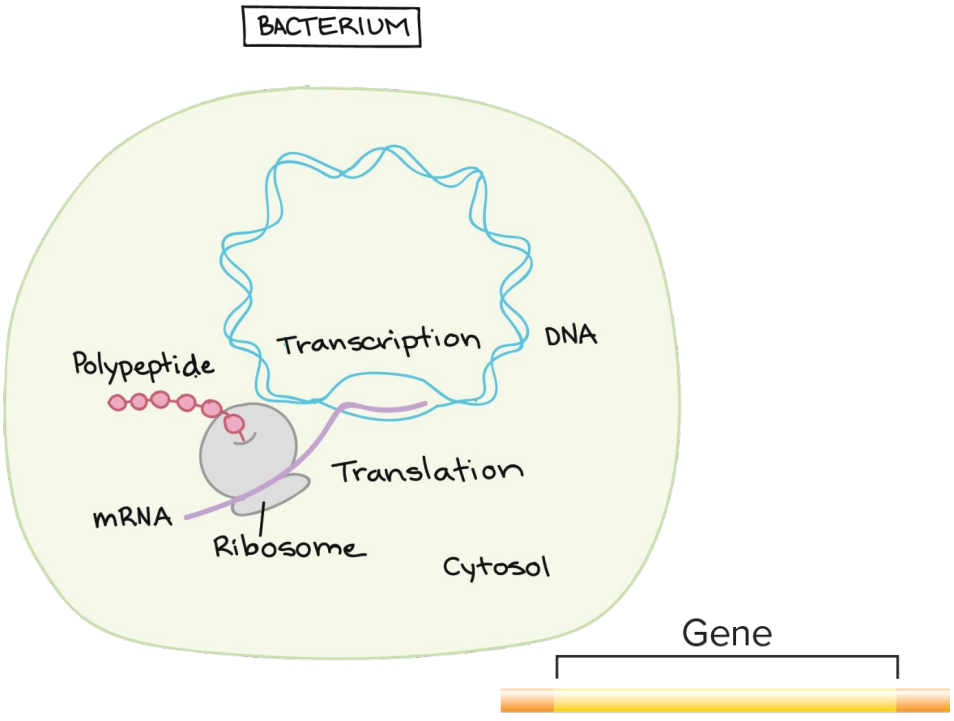


Eukaryotes extensively process their transcripts, whereas bacteria don't. Why?

1) Nuclear envelope separates transcription from translation

2) Introns separating exons need to be spliced out

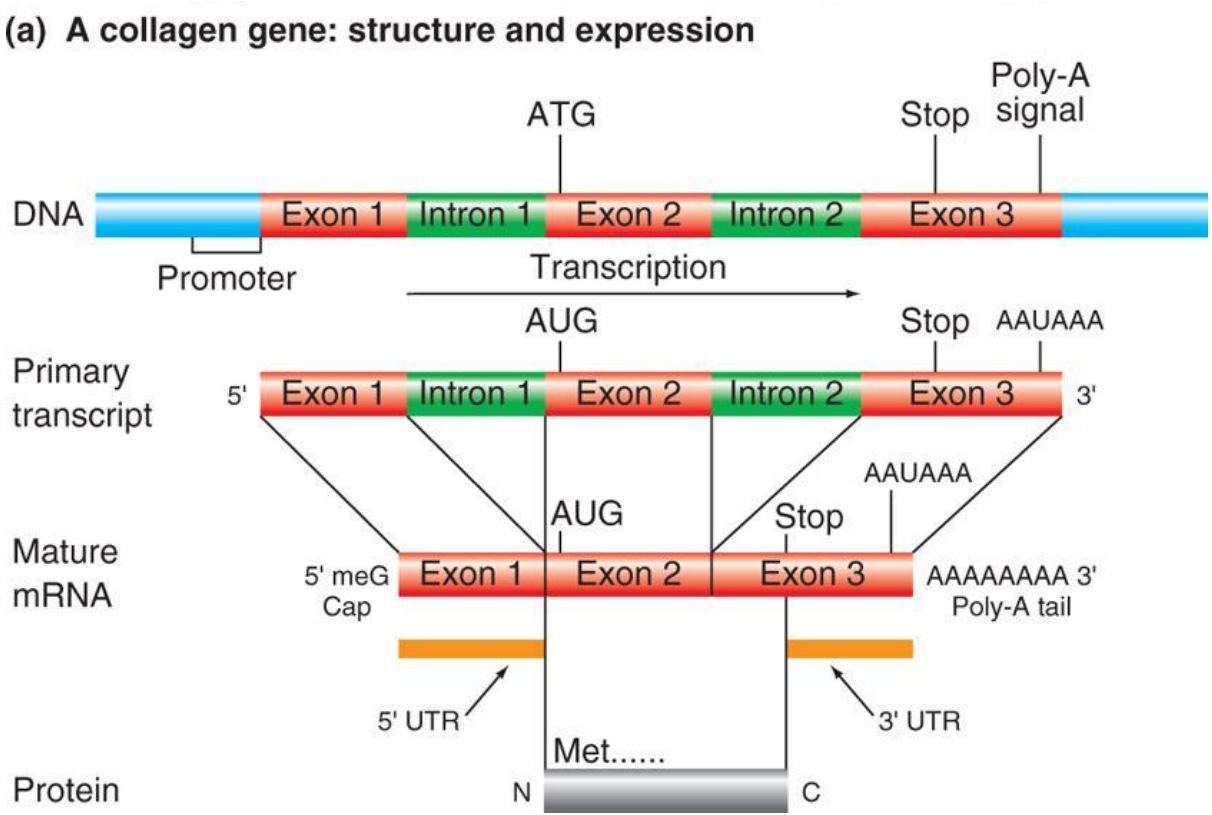




RNA splicing in eukaryotes

Notice that there is a 5’ UTR (untranslated region), then the coding region, then the 3’ UTR in the mature mRNA.

The coding region typically does not start at the very beginning of the transcript.



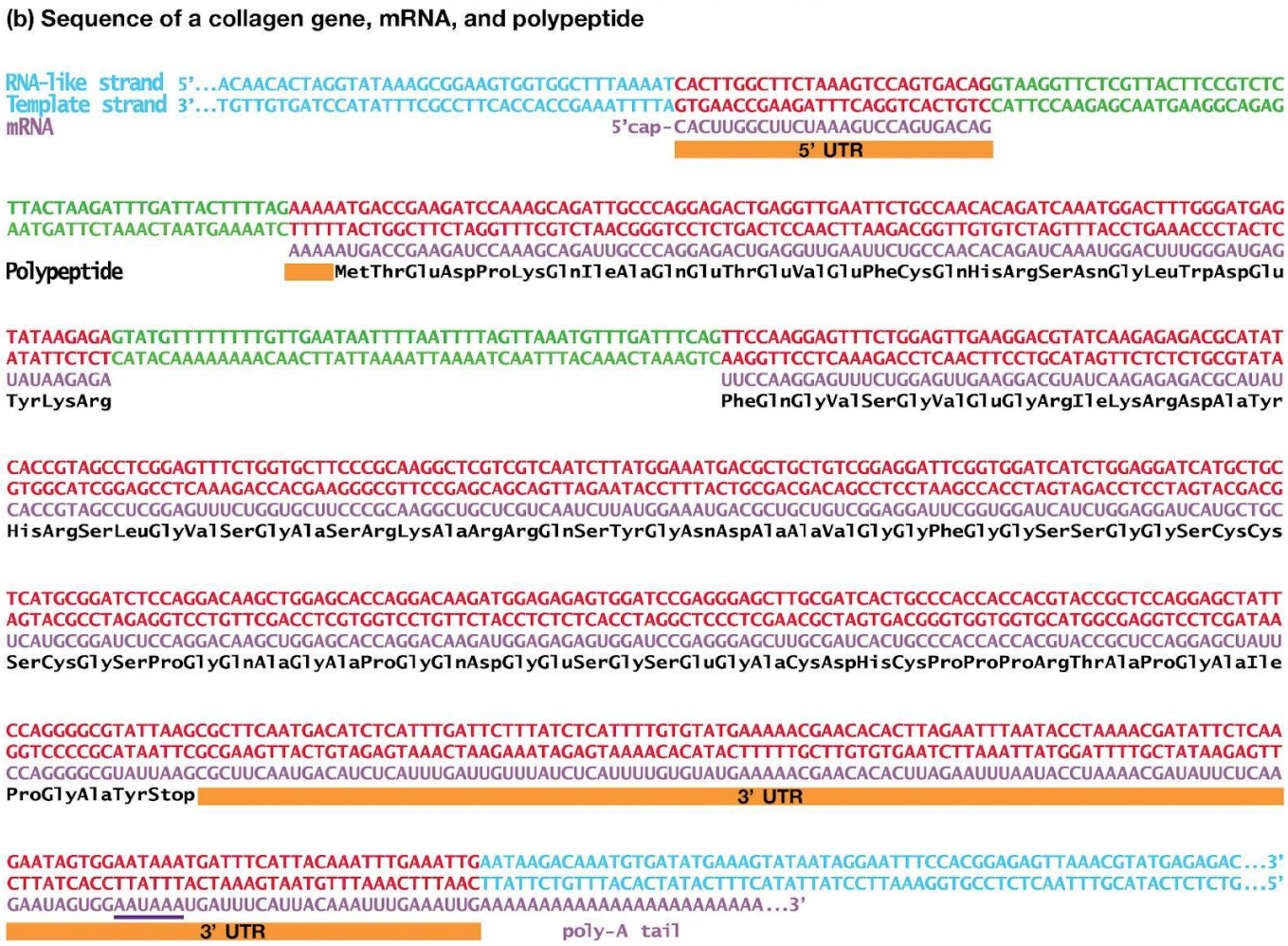
RNA splicing in eukaryotes

For eukaryotic genes, the full gene sequence contains the 5’ UTR, coding sequence, and 3’ UTR.

Introns (shown in green) are spliced out in the mature mRNA.

<u>Notice that introns can disrupt UTR or coding sequence.</u>

The 5’ cap and poly-A tail is added to the mature mRNA

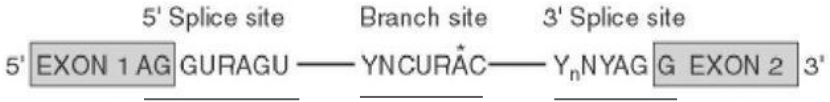


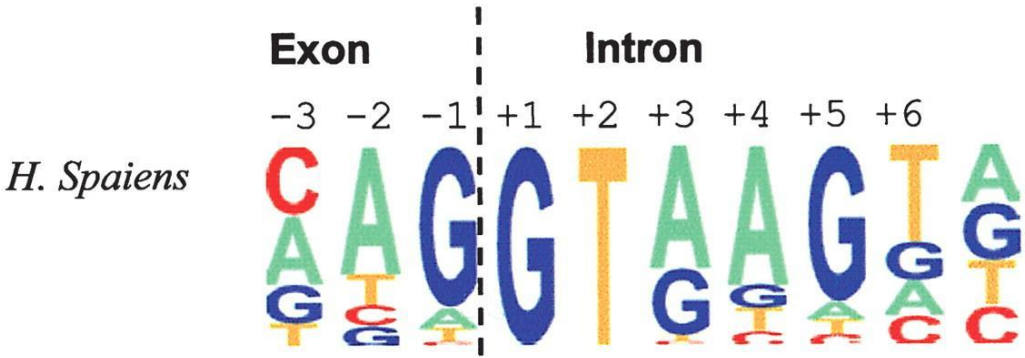
How does the cell know where the intron boundaries are?

5’ splice site, branch site, and 3’ splice site sequences are similar across all introns

Almost all introns **start with a GU** and **end with an AG**.

In RNA, **U** is used instead of **T.**

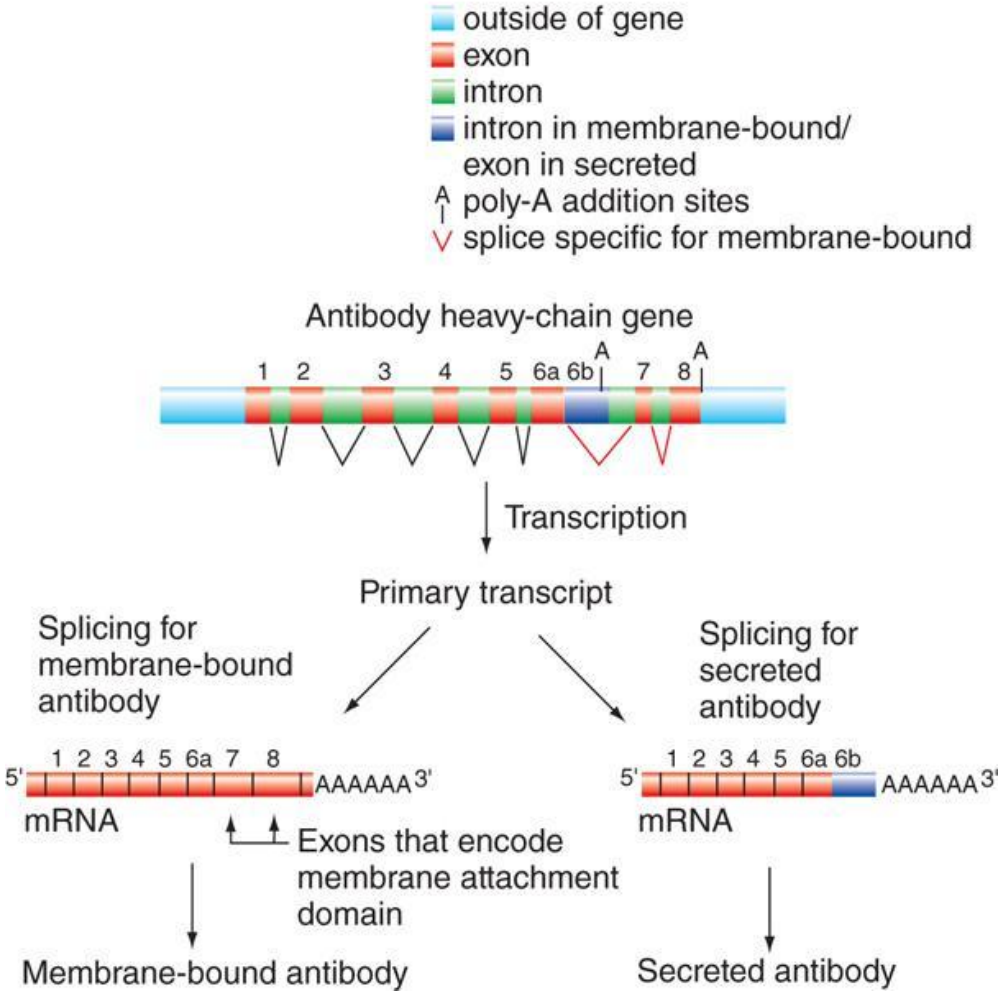




Alternative splicing

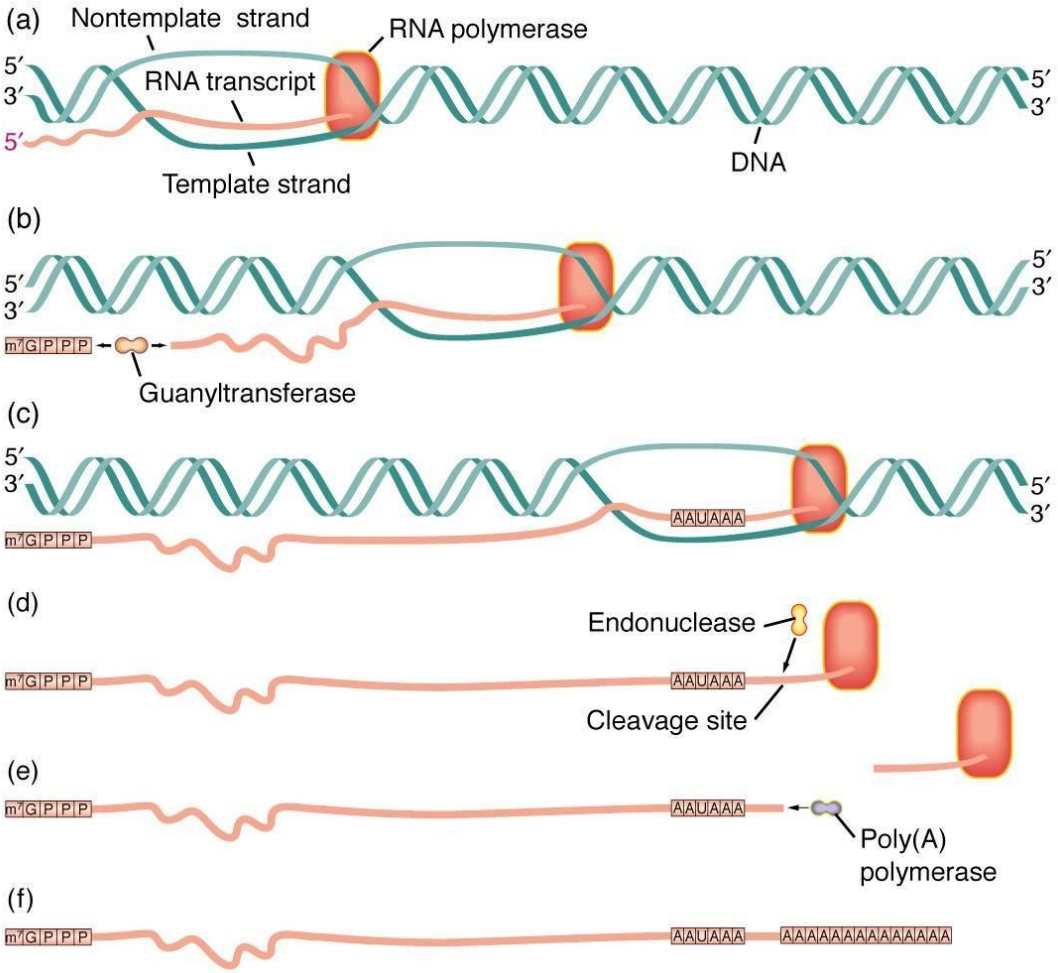
Over 95% of human genes are alternatively spliced.

Allows for one gene to produce many different gene products.



RNA processing in eukaryotes

- Requires:
  - 5’ methylated guanine cap
    - Guanylyltransferase (GTase) adds a guanosine monophosphate (GMP) group to the 5′-end of RNA
    - required for translation
  - Cleavage
  - Polyadenylation
    - Poly(A) polymerase tails mRNA
    - required for mRNA retention
- Bacteria mRNAs are not capped and do not have polyA tails



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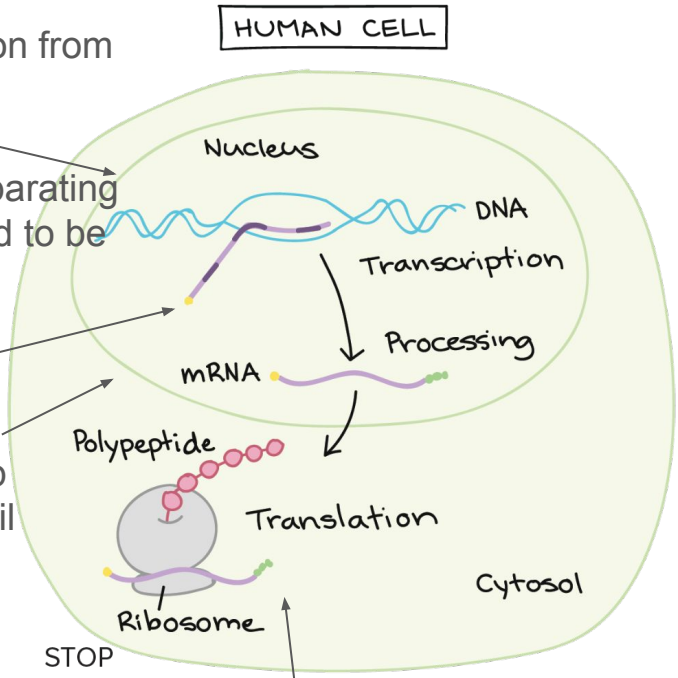
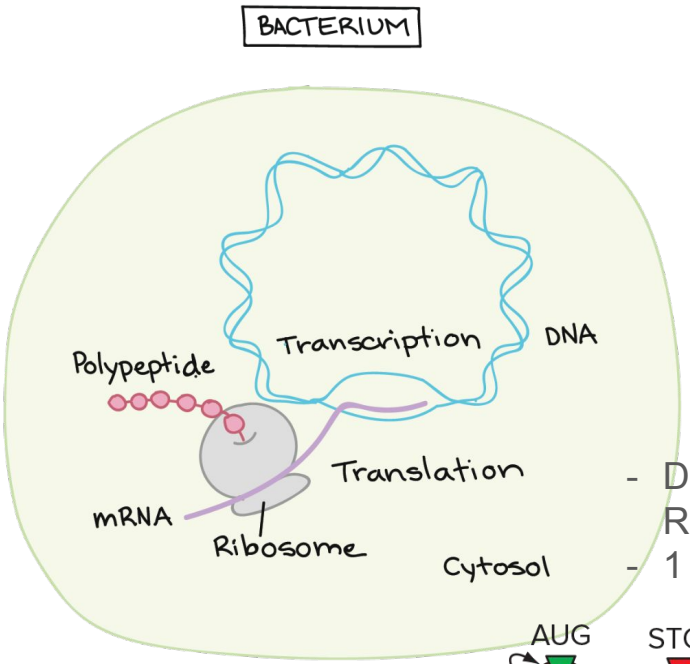
Eukaryotic transcript processing enables translation

1) Nuclear envelope separates transcription from translation

2) Introns separating exons need to be spliced out

3) Transcript 5'-CH3 cap and 3'-A tail must be added

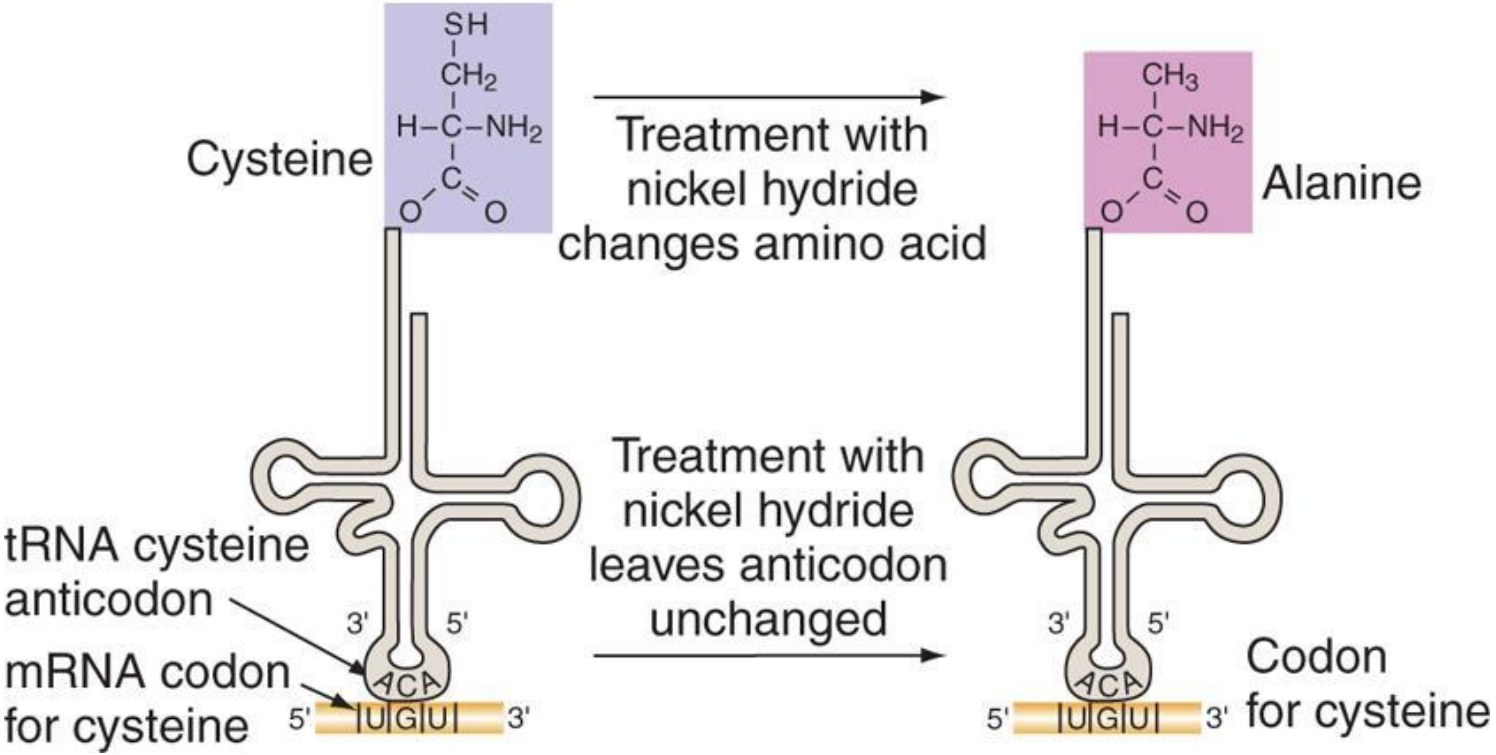




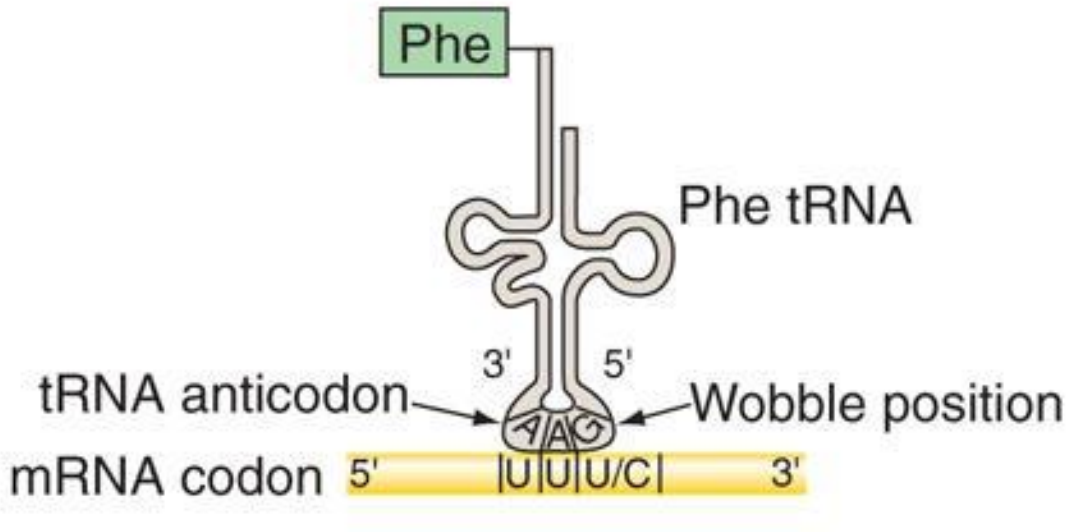
- Direct binding to RBS
- 1 mRNA, >1 RBS

4) Ribosome binds cap, then scans for RBS (1 mRNA, 1 RBS)

Base pairing between an mRNA codon and a tRNA anticodon determines what amino acid is added



Wobble position: position where base-pairing is more flexible



3rd codon position is degenerate

Mathematical degeneracy = "limiting case" where an object changes its nature to a simpler form.

The genetic code: codon to amino acid

Second Letter

1st letter	U		C		A		G		3rd letter
U	UUU UUC UUA UUG	Phe Leu Leu	UCU UCC UCA UCG	Ser Ser Ser	UAU UAC UAA UAG	Tyr Stop Stop	UGU UGC UGA UGG	Cys Stop Trp	U C A G
C	CUU CUC CUA CUG	Leu Leu Leu	CCU CCC CCA CCG	Pro Pro Pro	CAU CAC CAA CAG	His Gln Gln	CGU CGC CGA CGG	Arg Arg Arg	U C A G
A	AUU AUC AUA AUG	Ile Ile Met	ACU ACC ACA ACG	Thr Thr Thr	AAU AAC AAA AAG	Asn Lys Lys	AGU AGC AGA AGG	Ser Arg Arg	U C A G
G	GUU GUC GUA GUG	Val Val Val	GCU GCC GCA GCG	Ala Ala Ala	GAU GAC GAA GAG	Asp Glu Glu	GGU GGC GGA GGG	Gly Gly Gly	U C A G

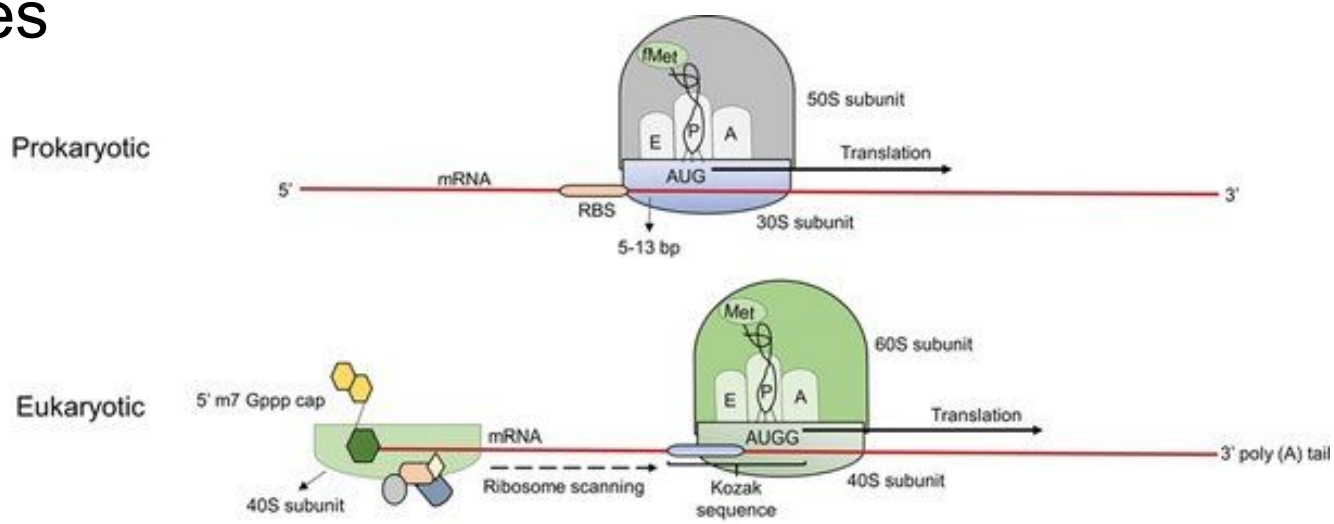
⇒ 3rd codon position is degenerate

Mathematical degeneracy = "limiting case" where an object changes its nature to a simpler form.

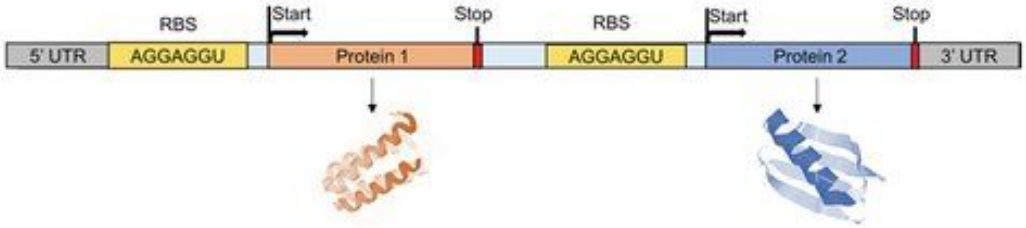
⇒ for many amino acids, any 3rd position nucleotide is equivalent

Translation initiation in eukaryotes vs bacteria: different binding sites

RBS ~ bacterial Shine-Dalgarno sequence



Eukaryotes have separate RBSs for ribosome binding and Kozak sequences for start codon recognition (overlaps start)



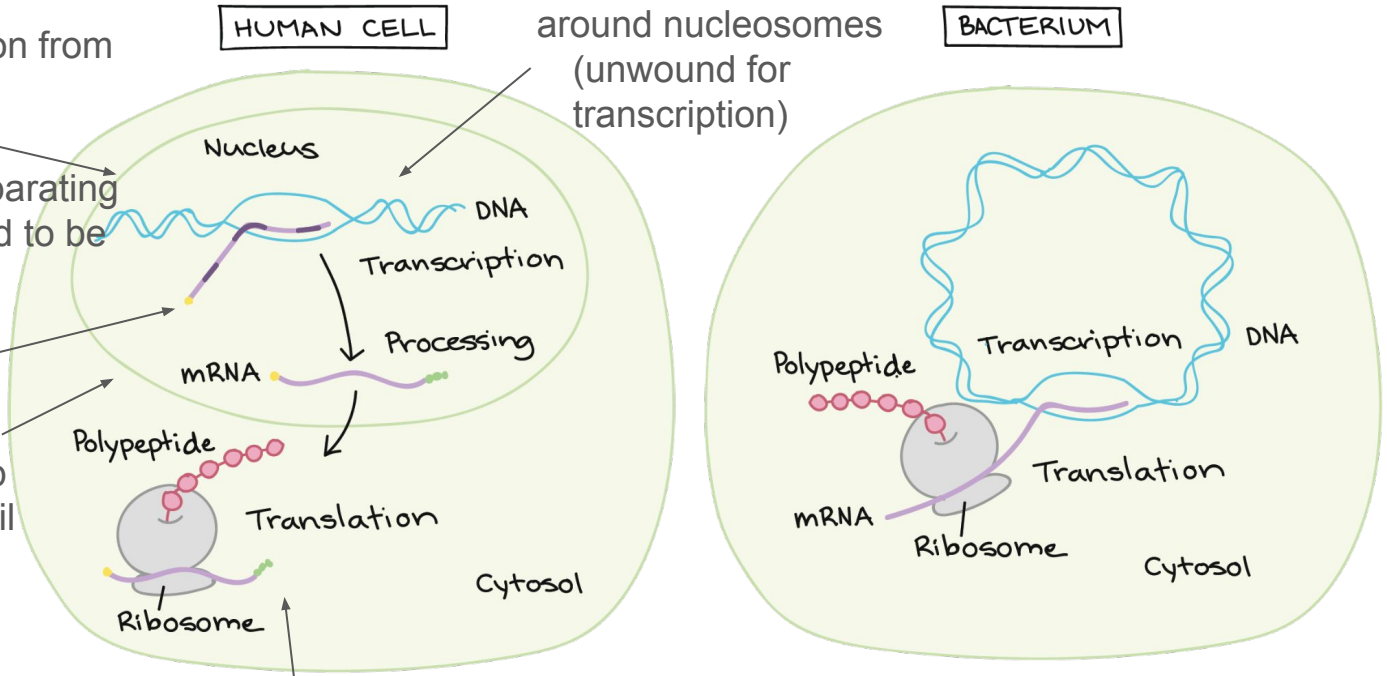
Eukaryotes extensively process their transcripts, whereas bacteria don't. They also have more mechanisms for regulating transcription.

1) Nuclear envelope separates transcription from translation

2) Introns separating exons need to be spliced out

3) Transcript 5'-CH3 cap and 3'-A tail must be added

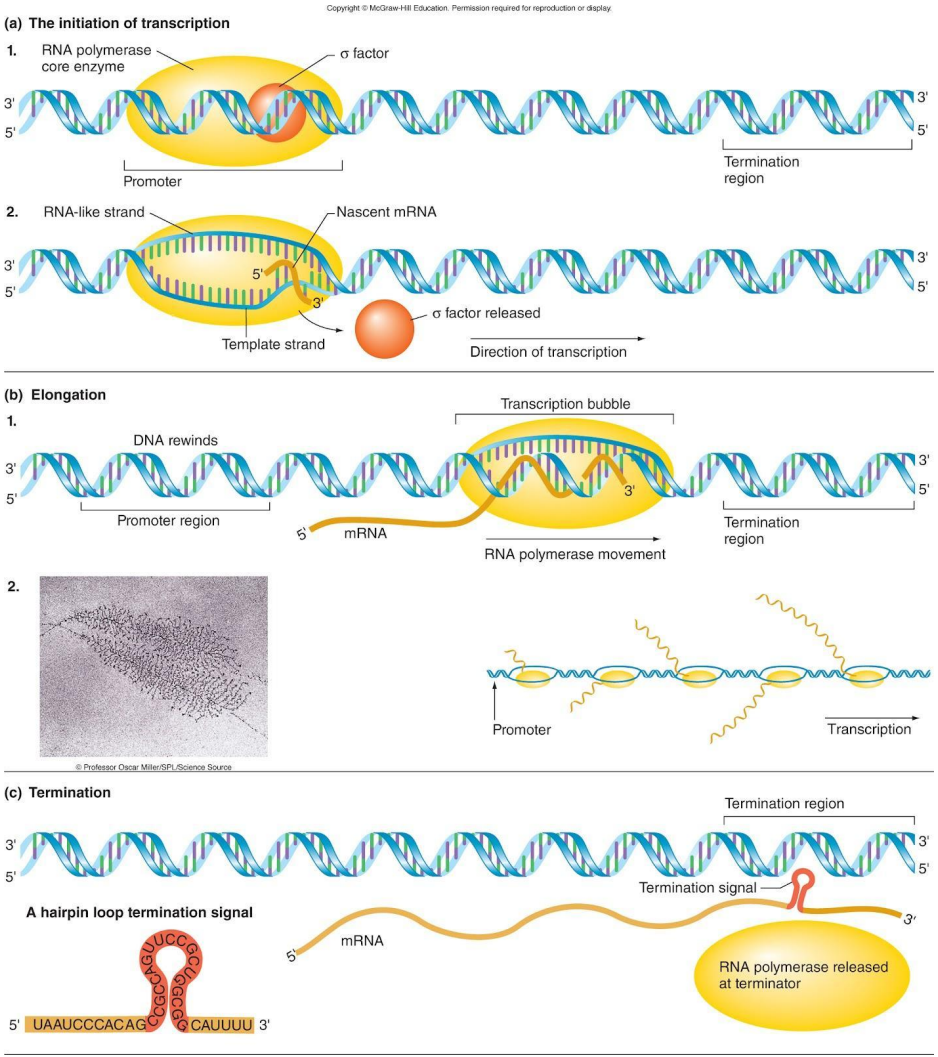
5) DNA is compacted around nucleosomes (unwound for transcription)



4) Ribosome binds cap, then scans for RBS (1 mRNA, 1 RBS)

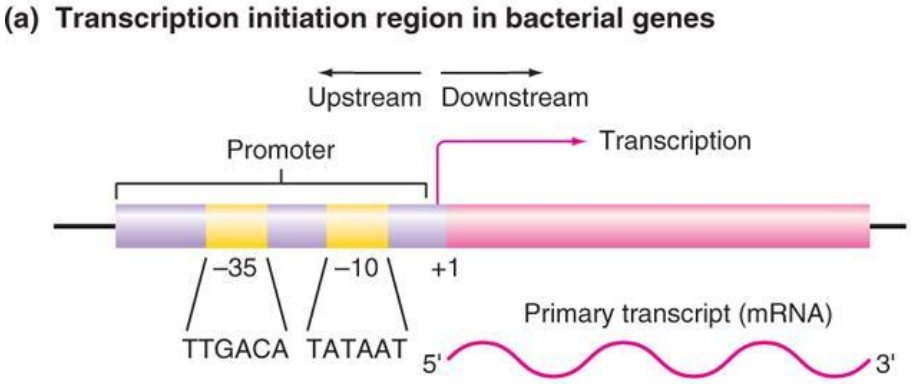
Transcription in prokaryotes (bacteria)

- Synthesized by RNA polymerase
- Initiated at promoters
  - Sequences recognized by transcription factors
  - On template strand
- The other strand is called the coding, RNA-like strand, or non-template strand
- Direct production of mRNA
  - No processing needed for translation
- Transcription termination signal induces hairpin that dislodges polymerase

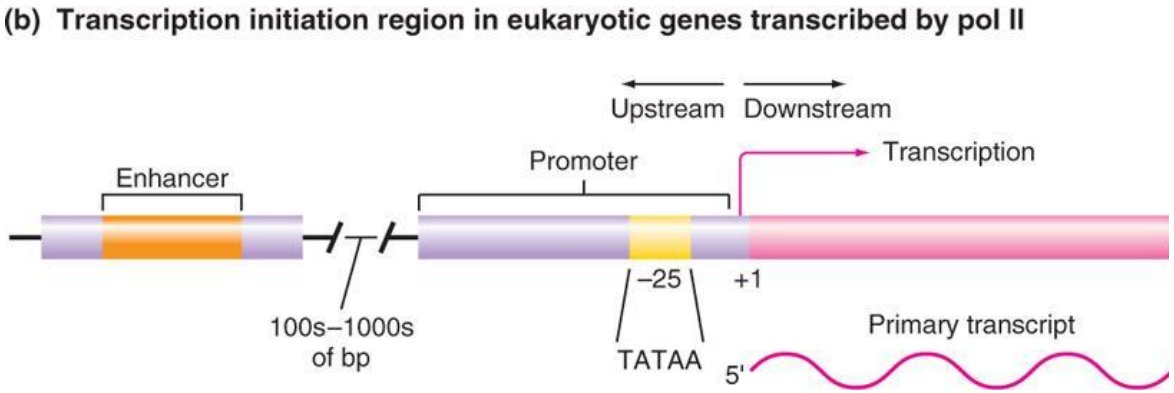


Differences in transcription control regions between bacteria vs eukaryotes

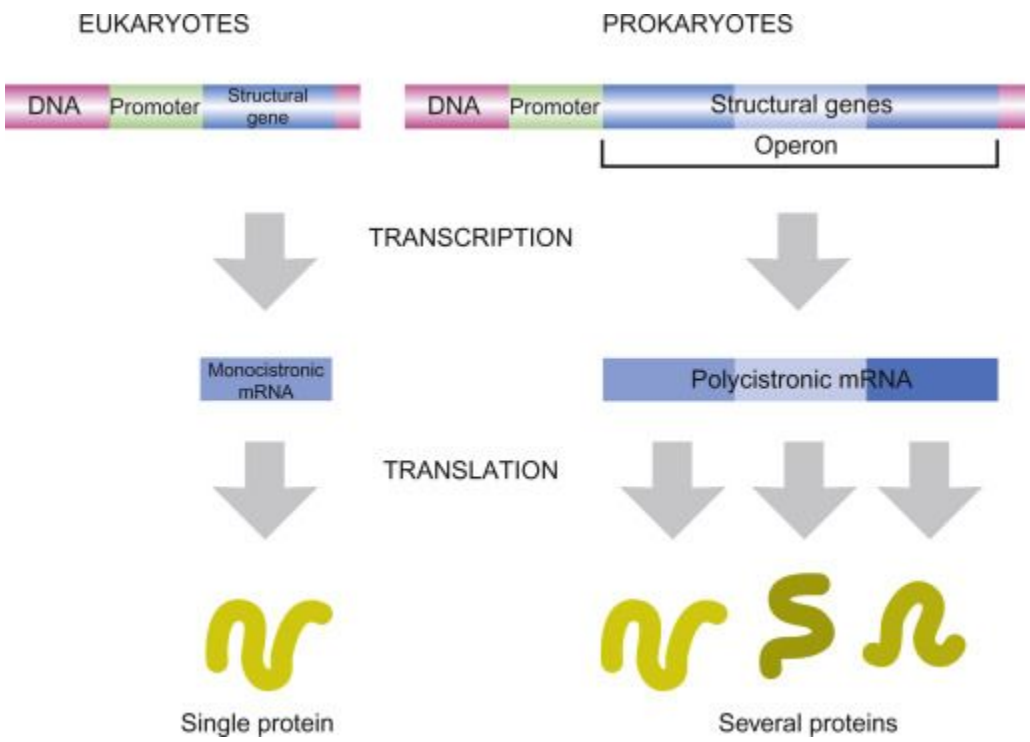
Regulatory regions are close by



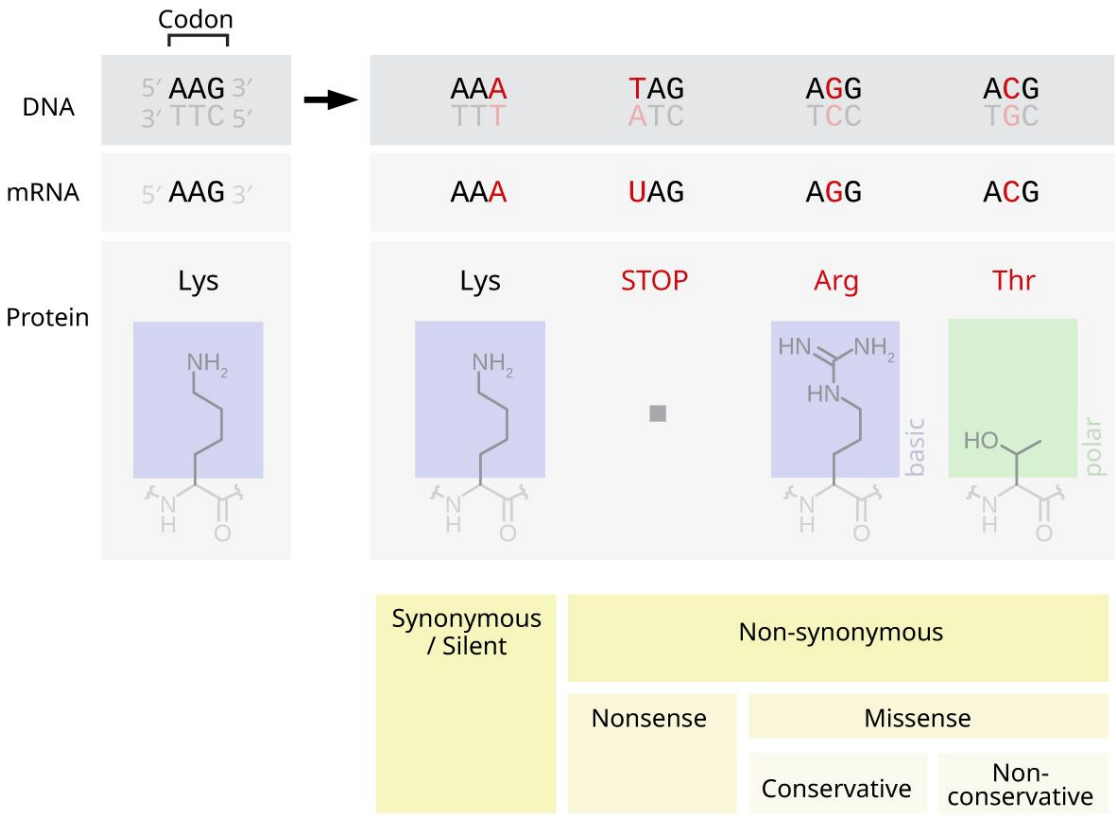
Regulatory regions are far away in the genome, and more numerous
(more on this in Lectures 14-15)



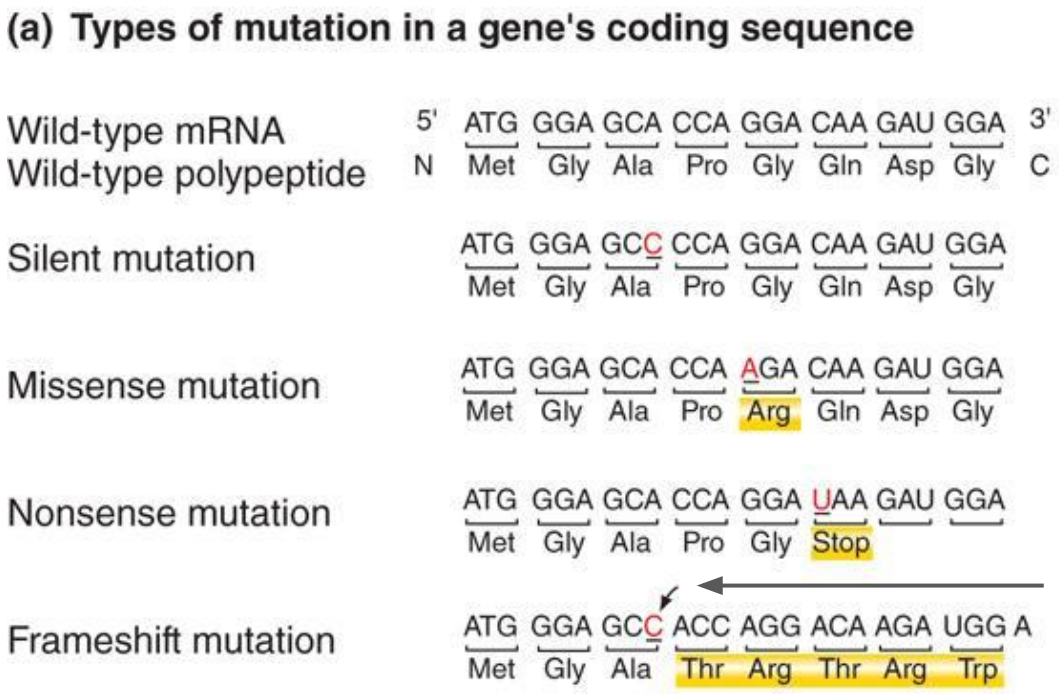
Polycistronic gene organization: simultaneous transcription and translation of more than one protein in bacteria



Mutations that affect the coding sequence

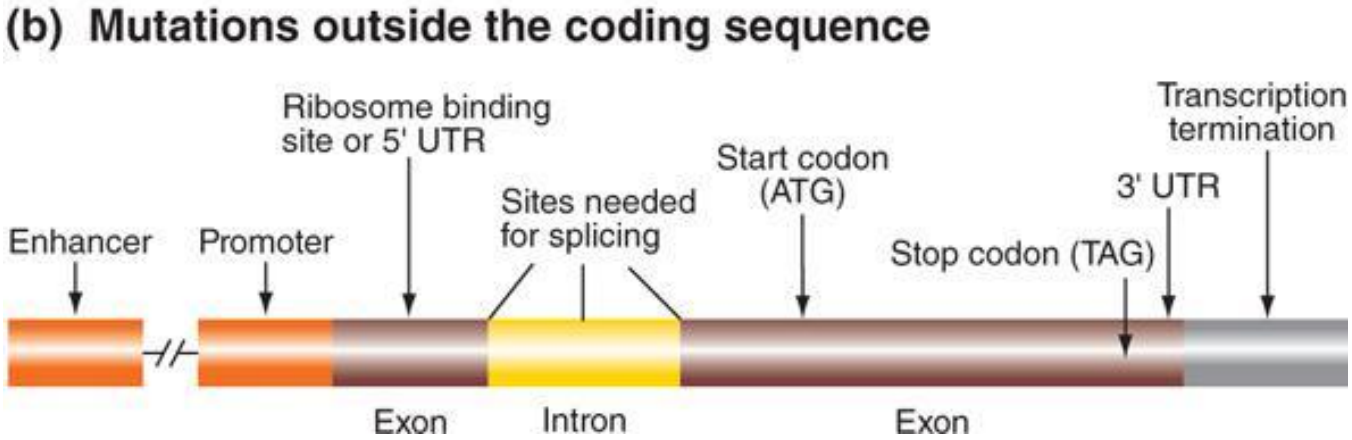


Mutations that affect the coding sequence



Inserted (or deleted) nucleotide shifts frame, *i.e.*, the unit of 3 base pairs that codons are read in

Mutations that affect non-coding sequence



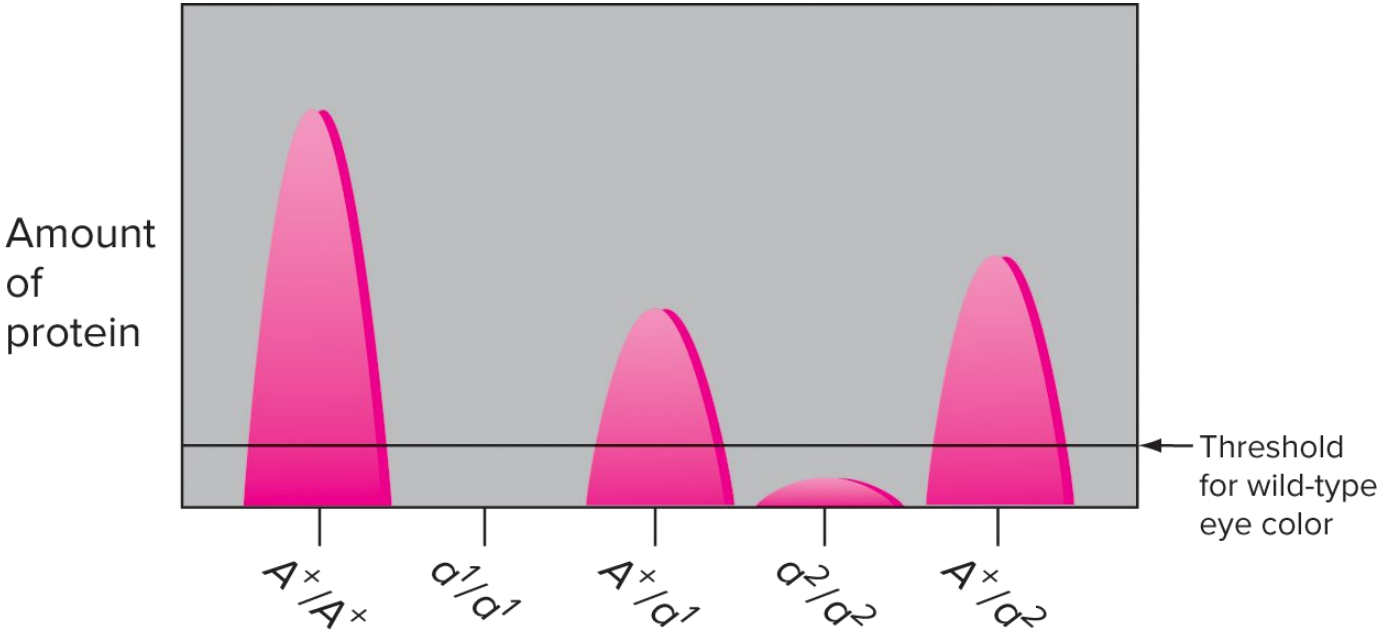
Most loss of function alleles are recessive to wild type alleles

Null or **amorphic** allele = no gene function, either:

- no protein expressed
- no functional protein

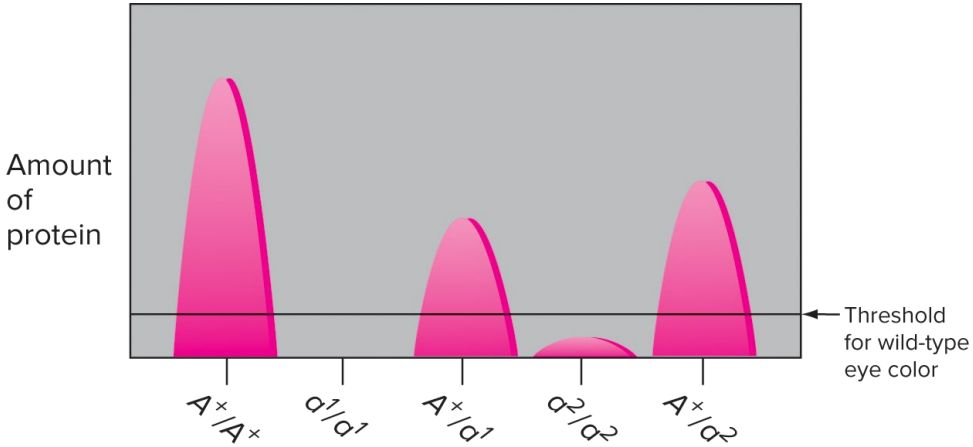
Hypomorphic allele = lower gene function, either

- Less protein expressed
- Less functional protein expressed

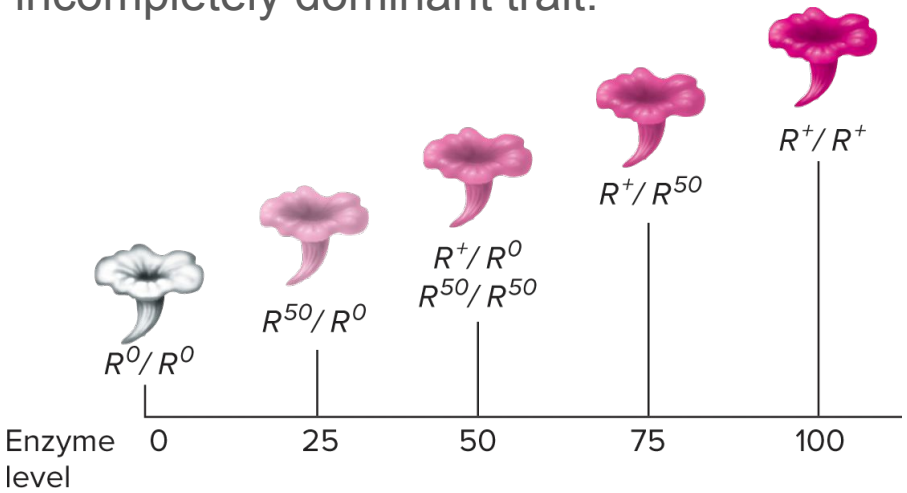


Incomplete dominance is the result of a phenotype varying continuously (opposed to discretely) with protein function

Discrete threshold of a completely dominant trait:



Continuous variation with an incompletely dominant trait:



Dominant loss of function alleles are haploinsufficient

Haploinsufficient = two functional alleles required for wild type phenotype

~800 haploinsufficient genes in human genome

Example:
GLI3+/GLI3 heterozygotes have polydactyly
GLI3 is a transcription factor

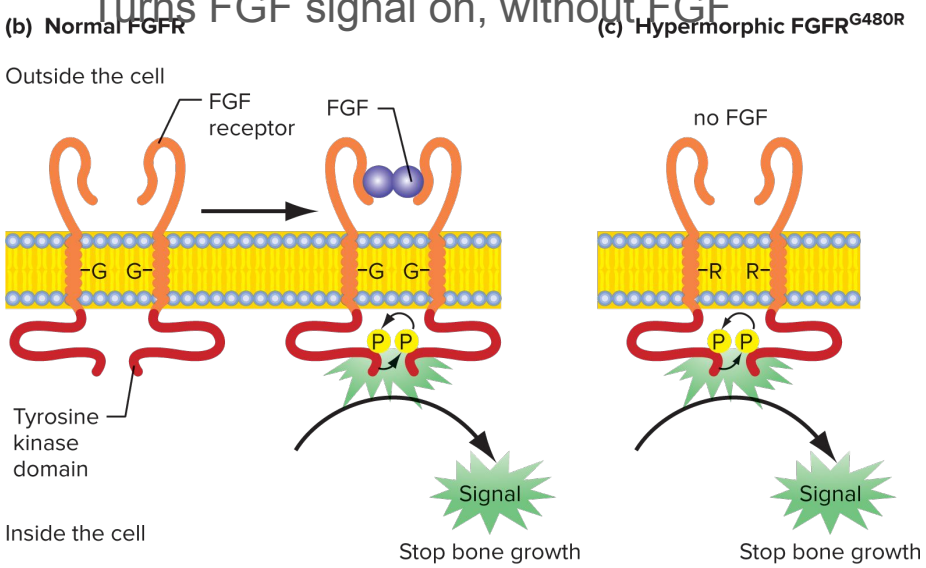
						643		
	V	T	K	K	Q	**R**	G	D
Normal allele ($GLI3^{+}$)	GTC	ACC	AAG	AAG	CAG	**C**GA	GGG	GAC
						↓		
Mutant allele (*GLI3*)	GTC	ACC	AAG	AAG	CAG	TGA	GGG	GAC
	V	T	K	K	**Q**	STOP		
					642			

Gain of function alleles are rare and almost always dominant

Hypermorphic alleles encode overactive proteins

E.g., FGFR3^{G480R} allele that causes achondroplasia

Turns FGF signal on, without FGF



(a) Achondroplasia

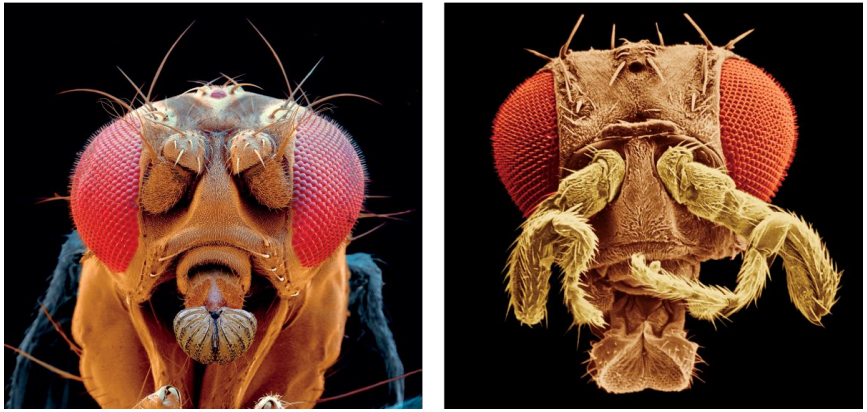
Gain of function alleles are rare and almost always dominant

Neomorphic alleles encode proteins that induce novel phenotypes

E.g., Antenna → legs in flies

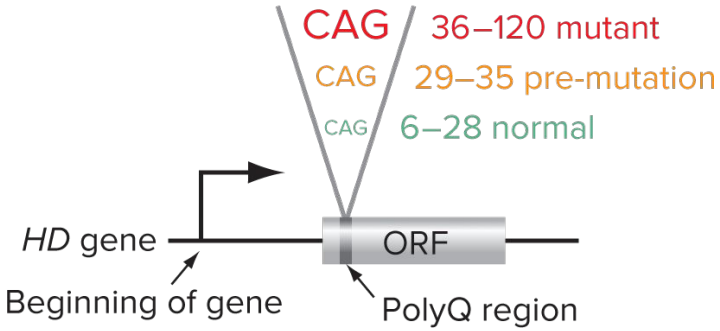
(a) The dominant neomorphic allele *Antp*^{Ns} causes antenna-to-leg transformation

Antp^{+}/*Antp*$^{+}$ | *Antp*^{Ns}/*Antp*$^{+}$



E.g., Huntington protein is toxic to nerve cells (wild type allele can't rescue effect)

PolyQ disease: Huntington disease



PolyQ = CAG glutamate repeat

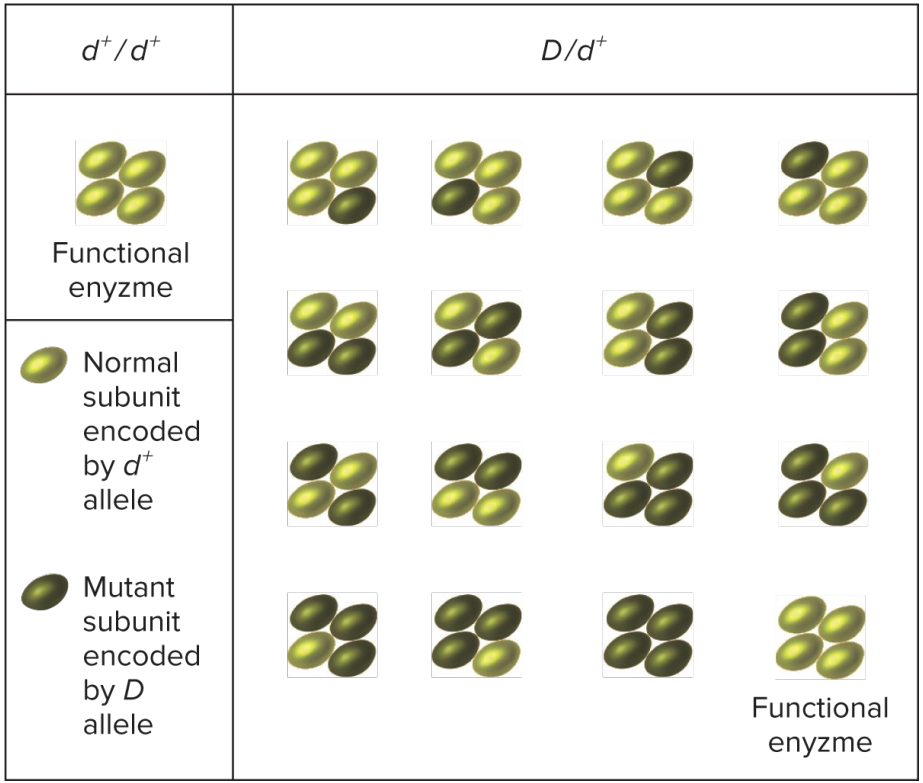
Dominant-negative alleles both fail to provide function and block wild type function

Antimorphic = Dominant-negative alleles

E.g., $Axin^{+}/Axin^{Kinky}$

(b) Phenotypic effect of the poison subunit encoded by the antimorphic allele *Axin*^{Kinky}

(a) The antimorphic allele *D* encodes a mutant poison subunit

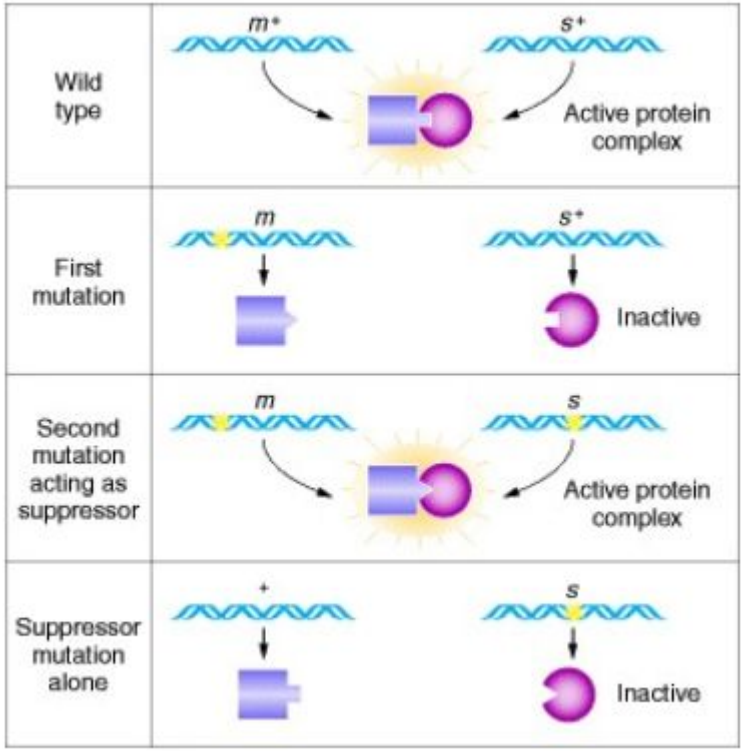


Another type of gene-gene interaction: suppression

A suppressor is an allele that reverses the effect of a mutation of another gene, resulting in the normal (wild-type) phenotype.

Both recessive and dominant suppressors are found, and they can act on recessive or dominant mutations.

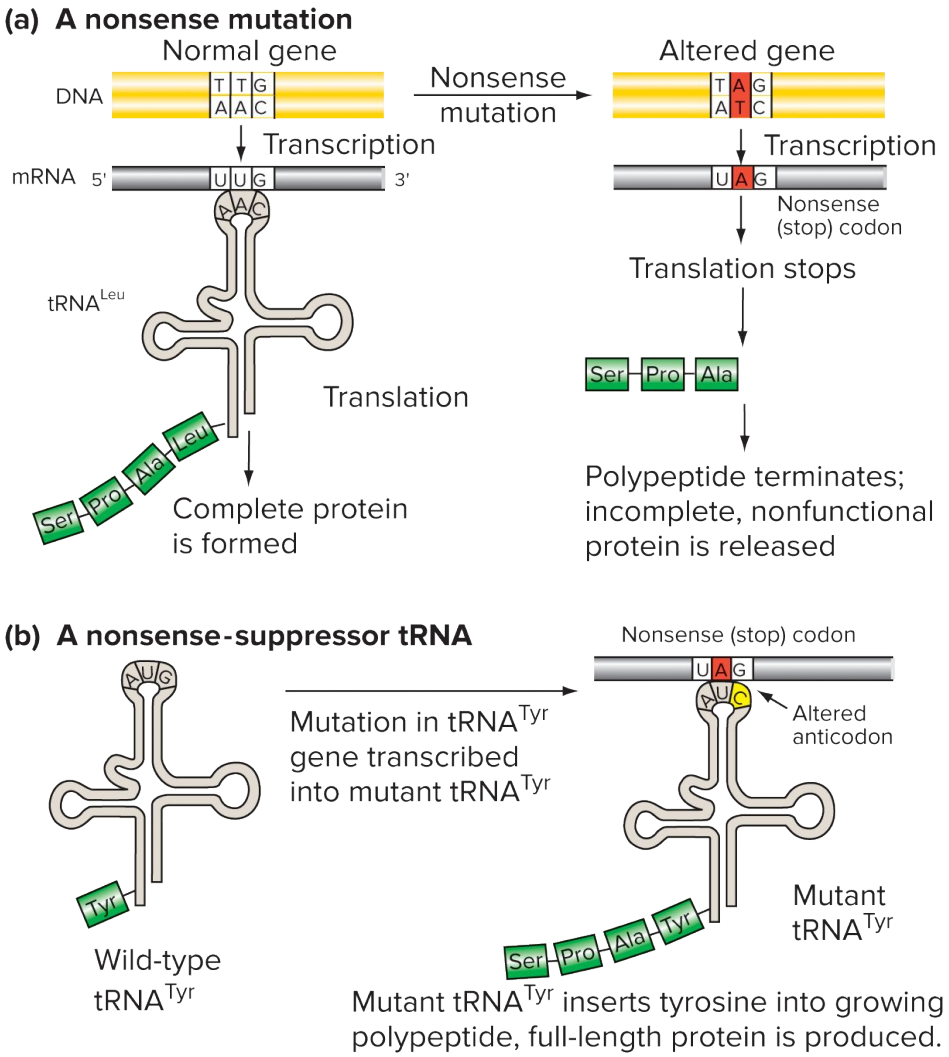
These possibilities result in a variety of different phenotypic ratios



Griffiths, et al. 2000

tRNA mutations can suppress stop codons

nonsense suppressor = tRNA with mutated anticodon inserts tyrosine at stop codons

Because tRNA genes are often present in several copies, a suppressor mutation in one of them will be perfectly viable.

Suppressor must be inefficient, otherwise too many proteins will be aberrantly long